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OBJECTIVE

A Mechatronics Engineering student aiming to specialize in Robotics Engineering and AI. I wish to learn and practice the various advances in research and development of Robotics Engineering, particularly in areas of Path Planning, Robot Perception and Multi-Robot Systems.

EDUCATION

Symbiosis Skills and Professional University, Pune **2019 - 2023**

Bachelor of Technology in Mechatronics Engineering *CGPA: 9.0337/10*

Relevant Coursework: Computer Programming, Control Engineering, Industrial Robotics, Theory of Mechanism, MATLAB and Simulink, Digital Signal Processing, Mechatronics System Design, Microprocessors and Applications.

EXPERIENCE

Technodune Pvt. Ltd. **June 2022-Dec 2022**

Robotics Engineering Intern *Pune, India*

- Optimized design solutions for Robots using solid and surface modelling in Solidworks and simultaneously worked with 3D printing (FDM) technology.
- Played a major role in developing software for two projects, "Wirelessly Controlled Bionic Arm" and "Swarm Robotics" using Python and C++ language.
- Worked closely with IoT framework and tools for applications in Robotics.

Symbiosis Skills and Professional University **Mar 2022 – May 2022**

Undergraduate Research Assistant *Pune, India*

- Worked as a research assistant for two papers titled, "Wearable Glove for Sign Language to Regional Language Conversion" (paper accepted for presentation in IEEE conference) and "Sign Language Recognition using Deep learning" (paper under review).
- Trained the ML model with the help of a LSTM neural network using landmarks of hand-poses obtained through MP Holistic.
- Tested both the glove and ML model for a set of hand gestures for Marathi (native language) letters/words and achieved an accuracy in sign recognition ranging from **85%** - **97%**.

Automate India Pvt. Ltd **Aug 2021 – Sept 2021**

Mechatronics Engineering Intern *Pune, India*

- Assisted in developing auto-clutch kits consisting of a microcontroller, sensors and actuator and tested the kits in manual cars to automate movement of clutch.
- Studied the program of microcontroller, written in C, to control movement of clutch based on feedback received by clutch position sensor in conditions of variable speed and brake.
- Designed various parts required for actuator assembly such as mounting plates, sensor housings etc. in Fusion 360.

PROJECTS

Real-Time Robot Detection using OpenCV **Sept 2022 - Present**

Technodune Pvt. Ltd.

- Formulated a perspective correction algorithm to straighten frames obtained through real-time video and to maintain accuracy of Robot coordinates compared to real-world position.
- Implemented **Robot Pose Estimation** using two techniques:
    1. Detection of ArUco markers placed on the robot.
    2. Detection using YoloV5 and PyTorch
- Achieved an accuracy of **85-90%** in robot detection
- Designed enclosure of the mobile omni-wheeled robot in Solidworks.

Autonomous Mobile Robot (ROS Mentorship Program) **July 2022 - Oct 2022**

RigBetel Labs

- Designed a differential wheeled robot in Fusion 360 and implemented its URDF conversion.
- Modified the publisher and subscriber ROS nodes as per the requirements of the robot.
- Implemented **autonomous navigation** using **SLAM** algorithm in Gazebo simulation environment.

Wirelessly Controlled Bionic Arm **June 2022 - Aug 2022**

Technodune Pvt. Ltd

- Designed a "position-controlled" **7DOF robotic arm** intended to be used for on-desk applications, controlled by hand gestures using a glove in addition to a customised webpage that allows performing different tasks with the arm.
- Programmed the **ESP32S** for data transfer between the arm and the glove (and webpage) using WebSocket protocol.
- Programmed the **ATMega** microcontroller for implementing robot manipulator kinematics.

Sensor-Based Glove for Indian Sign Language Recognition **Mar 2022 - May 2022**

Symbiosis Skills and Professional University

- Developed a prototype of a **sensor-based glove** which detects the hand gestures used in sign language through data provided by low power 3-axis accelerometers and provides a speech output post conversion to text.
- Set threshold acceleration values obtained for each of the signs through trials to optimize recognition of sign language and achieved an accuracy of **91%** in sign detection.
- Interfaced the on-board **ATMega** microcontroller with a speaker to execute text-to-speech conversion. Tested the audio output for native language (Marathi).

TECHNICAL PAPERS

- **Paper Title:** "Marathi Sign Language Hand Gesture Recognition Using Accelerometer and 3D Printed Gloves"
  Status: Published in IEEE proceedings of International Conference on Computational Intelligence and Communication Networks (CICN) held on Dec 4, 2022 [Paper]

SKILLS

- **Programming Languages**: C++, C, Python
- **Libraries, Tools and Software**: ROS(Noetic), ROS 2 (Humble), Gazebo, RViz, OpenCV, NumPy, Matplotlib, PyTorch, MATLAB, Fusion 360, Solidworks, Ultimaker Cura.

CO-CURRICULAR ACTIVITIES

Robotics and Automation Training (Certification) **July 2021**

FESTO India Pvt. Ltd.

- Simulated operations of various industrial pneumatic and electro-pneumatic control systems.
- Implemented PLC programming in CODESYS software for industrial robotics simulation.
- Worked on 3D Virtual Training platform- CIROS for Industrial Robot programming and computer integrated robotic control.

Internship Coordinator **Aug 2019 – May 2021**

Symbiosis Skills and Professional University

- Conversed with students regarding their technical interests and skills.
- Communicated with industry professionals about their expectations from interns.
- Organized on-campus interviews and drives as a part of the mandatory summer internship program of the university.